

MAURIZIO VASSALLO

Data Scientist · AI Researcher

Curious and multidisciplinary researcher focused on applying AI across different domains. My work applies data science and machine learning to solve real-world problems in areas like optimization and intelligent decision-making.

PERSONAL INFO

- 🇮🇹 Italian
- ✉️ mauri.vass.dev@gmail.com
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REWARDS

- Awarded a competitive research grant (€2,300) to fund Master's thesis development in Belgium.
- Achieved 2nd place in an Artificial Intelligence competition, securing a cash prize (€500) among numerous participants.
- Won full scholarship (€6,500) for an exchange year at Tongji University (同济大学), ranking in the top 21% of applicants.

LANGUAGES

- Italian:** Native
- English:** Advanced (C1)
- French:** Intermediate (B2)
- Chinese:** Intermediate (HSK 3)

SKILLS

- Programming Languages:** Python, C/C++, C#, Java, JavaScript, MATLAB
- Machine Learning & Data:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, Spark, SQL, Hadoop
- Data Visualization:** Matplotlib, Seaborn, Plotly
- Development Tools:** Git, Docker, L^AT_EX
- Systems & Software:** Linux, Windows, Unity3D, Unreal Engine, Blender, QGIS, Photo-shop
- Web & Protocols:** REST APIs, MQTT, HTML, CSS

EDUCATION

- 10/2022 - **Ph.D. student: AI in Power Distribution Networks**
GOING **University of Liège** Liège, Belgium
- 10/2019 - **Master's degree: Data Science and Engineering**
07/2022 **Polytechnic of Turin** Turin, Italy
- 09/2017 - **Bachelor's degree: Computer Science Engineering**
12/2021 **Tongji University (同济大学)** Shanghai, China
- 09/2016 - **Bachelor's degree: Computer Science Engineering**
09/2019 **Polytechnic of Turin** Turin, Italy

EXPERIENCES

- 02/2022 - **Overvoltage Situations Forecast & Active Control in a Medium-Voltage Power Grid**
06/2022 **M.Sc. Thesis | University of Liège** Liège, Belgium
Compared different ML models to forecast overvoltage risks in a real medium-voltage grid. Implemented an RL agent to control generators, successfully mitigating voltage violations with minimal energy curtailment. (Github)
- 09/2021 - **Collision Avoidance System using Reinforcement Learning**
12/2021 **B.Sc. Thesis | Tongji University (同济大学)** Shanghai, China
Researched and implemented RL architectures in a simulated UE4 environment. The agents learned to navigate complex tracks using LiDAR data, demonstrating zero-shot transfer to unseen environments. (Github)
- 04/2021 - **AI & Backend Developer**
07/2021 **Internship | Morecognition** Turin, Italy
Engineered an ML pipeline to classify complex upper-limb movements from multi-sensor IMU/EMG data. Built and deployed a scalable backend server to manage user data and exercise analytics, supporting the development of a physiotherapy analytics platform. (Github)
- 03/2021 - **New Opportunities from Autonomous and Connected Mobility**
06/2021 **Innovation Challenge | CLIK** Turin, Italy
Developed a V2X (Vehicle-to-Everything) communication prototype that delivered real-time parking availability to drivers, improving traffic flow and reducing search time in urban areas.
- 04/2019 - **Java Junior Developer**
06/2019 **Internship | INTESA** Turin, Italy
Developed and deployed a new company website, optimizing SQL queries to improve database response time. Gained full-stack experience contributing to both back-end (Java) and front-end (JavaScript, HTML, CSS) components.
- 09/2019 - **Cybersecurity Academy**
10/2019 **Professional Training | Randstad** Milan, Italy
Intensive training in cybersecurity essentials including threat analysis, vulnerability assessment, security protocols, and defensive coding practices across hardware and software systems.
- 07/2017 **The art of Game Programming**
Summer School | Artifex Mundi Gliwice, Poland
Intensive summer program on Unity3D, covering fundamental principles of game physics, graphics, and system mechanics.